

# TITANIC DATA ANALYSIS

# INTRODUCTION

This project analyzes passengers' survival patterns using MySQL and Power BI. The project focuses on how survival varied across gender, embarkment port, age group and passenger class.

The purpose of this analysis is to identify patterns and differences in survival rates of passengers.

# RESEARCH QUESTIONS

1. How were survivors distributed by gender?
2. How were survivors distributed by embarkation port?
3. How were the survivors distributed across each group?
4. How did survival rates vary across passenger classes?

# DATASET OVERVIEW

## **Dataset:**

- The dataset consists of 714 rows.
- Main variables – Sex, Age, PClass, Embarked, Survived

## **Methodology:**

- MySQL: Used for data exploration, statistical calculations and comparison.
- Power BI: Used to create interactive data visualization and explore trends and patterns.

# ANALYSIS AND CALCULATION

1. How were survivors distributed by gender?

The analysis shows that female passengers had a higher survival rate compared to males on the ship.

2. How were survivors distributed by embarkation port?

Passengers who embarked from Southampton had a highest survival rate followed by Cherbourg while the lowest survival rates were from Queenstown.

3. How were the survivors distributed across each group?

Passengers between the age group of 0 to 17 (Children) had highest survival rates while older passengers above 60 (Elderly) had lowest rate of survival.

4. How did survival rates vary across passenger classes?

Survivors from 1st class had the highest survival rate followed by 2nd class while 3rd class had lowest survival rate out of the three classes.

# KEY FINDINGS

- Female passengers had higher survival rate
- Passengers embarking from Southampton were recorded with the highest survival rate while passengers embarking from Queenstown recorded the lowest survival rate.
- Children had the highest survival rate while elderly had the lowest survival rate.
- Passengers from 1st class had the highest survival rate while 3rd class passengers had lowest survival rate.

# CONCLUSION

The analysis compared survival rates across gender, embarkation port, age group, and passenger class. Patterns were observed across these passenger characteristics, providing insight into how survival varied among different groups within the dataset.

Overall, the project demonstrates the use of SQL for data aggregation and analysis and Power BI for visualizing and communicating findings.